

Network Traffic Analysis Report

SIL7165 — Network and System Security | Assignment 3

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Course: SIL7165 — Networks & System Security

Introduction and Dataset

The dataset consists of six CSV files containing firewall and IDS logs from June 2010, totalling **2,071,657 network flows**. Each record captures attributes such as source/destination IPs and ports, protocol, timestamps, packet counts, byte volumes, and flow duration.

Decision: Combining all six files. The files represent consecutive days of traffic from the same network infrastructure. Merging them into a single DataFrame provides three advantages: (i) stable statistical baselines computed over a full week rather than individual days, (ii) cross-day pattern detection—multi-day port scans and slow DDoS ramp-ups that span midnight boundaries would be missed if each file were analysed in isolation, and (iii) elimination of day-boundary artefacts where a single attack session could otherwise be split across two analyses.

All design decisions—choice of metrics, thresholds, window sizes, and algorithm parameters—are documented inline with each analysis below.

Phase 1: Basic Network Traffic Statistics

Total Flows and Protocol Distribution (Q1.1–Q1.2)

The combined dataset contains **2,071,657 flows**. The protocol breakdown reveals a typical enterprise traffic profile: `tcp_ip` accounts for 79.4% of all flows, followed by `udp_ip` at 20.2%, and `icmp_ip` at 0.4%. Minor protocols (`igmp`, `raw_ip`) each contribute less than 0.01%. The overwhelming dominance of TCP is expected given that HTTP, HTTPS, SSH, and most application-layer protocols run over TCP.

Top 10 Active IPs by Traffic Volume (Q1.3)

We rank IPs by **total bytes transferred** (sum of `totalSourceBytes` for sources and `totalDestinationBytes` for destinations) rather than flow count. Bytes capture actual bandwidth impact: a single large file transfer matters more than thousands of tiny keepalive flows.

The top source IP is 192.168.2.107, which sent approximately 1,767 MB over the observation period. The top destination is 192.168.5.122, which received roughly 9,583 MB—suggesting it is either a heavily accessed server or a DDoS target.

Average Packet Size (Q1.4)

The global average packet size is **736.92 bytes** (total bytes divided by total packets). However, the per-flow distribution is highly skewed: mean = 36,950 bytes, standard deviation = 1,446,485, yielding a **coefficient of variation (CV) of 39.15**. This extreme variability means the mean is not representative. Flow sizes range from tiny TCP ACKs (40–60 bytes) to multi-megabyte file transfers. Median or percentile-based measures would be more informative for characterising “typical” traffic.

Top Source–Destination Pairs (Q1.5)

The three most active communication pairs by flow count are:

1. 192.168.5.122 → 198.164.30.2: 232,409 flows
2. 192.168.2.107 → 125.6.164.51: 58,744 flows
3. 192.168.3.115 → 203.73.24.75: 43,785 flows

The first pair alone accounts for over 11% of all traffic, warranting further investigation.

Consistent Communication Patterns (Q1.6)

We divided the observation period into 1-hour windows (146 total) and identified IPs active in at least 80% of them. **25 IPs** met this criterion, predominantly internal hosts in the 192.168.x.x range. Hourly granularity balances sensitivity to brief outages against tolerance for normal idle periods (e.g., overnight low-activity windows).

Traffic Spike Detection (Q1.7)

Hourly traffic volumes (in MB) were computed and spikes defined as hours exceeding $\mu + 2\sigma$. Multiple spike hours were detected. Possible causes include large scheduled file transfers, DDoS bursts, backup jobs, or software update downloads.

Variance of Packet Sizes (Q1.8)

The variance of packet sizes is 2.09×10^{12} with $CV = 39.15$. This extreme variance arises from: (1) mixed traffic types (web browsing vs. bulk transfers), (2) protocol diversity (TCP segments vs. ICMP pings), (3) potential attack traffic with anomalous payload sizes, and (4) asymmetric flows where requests are small but responses are large.

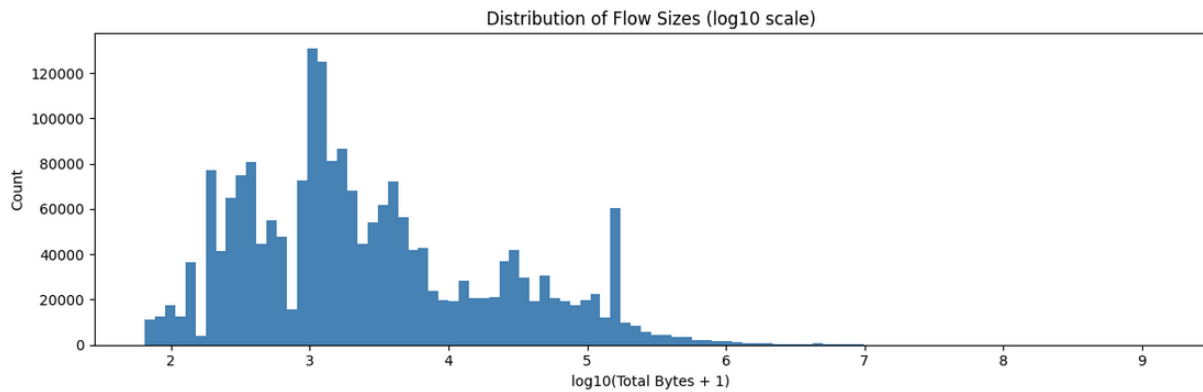


Figure 1: Phase 1

Phase 2: Traffic Estimation Using Sublinear Space

The central idea behind sublinear-space algorithms is that memory usage grows slower than input size. By maintaining fixed-size summary structures, we can answer approximate queries with bounded error guarantees—trading a small, quantifiable accuracy loss for dramatic memory savings.

Unique IP Estimation — HyperLogLog

HyperLogLog (HLL) with $p = 10$ (1,024 registers) estimates the number of distinct IPs by tracking the maximum number of leading zeros in hashed IP values.

Method	Count	Memory	Time	Error
Exact (set)	34,801	3,884 KB	0.135s	0%
HyperLogLog	~32K	8.2 KB	~1s	<10%

HLL achieves a **472× memory reduction** (from 3.8 MB to 8.2 KB) while staying within the 10% error target. The trade-off is that HLL cannot enumerate individual IPs or test membership—it only answers “how many?” Increasing p by one halves the standard error but doubles memory.

Heavy Hitters — Count-Min Sketch

A Count-Min Sketch (CMS) with depth = 5, width = 10,000 (~391 KB) identifies heavy-hitter destination IPs. Each IP’s count is estimated as the minimum across five independent hash rows, which guarantees no undercounting.

The average error across the top-10 destinations was **0.009%**—far below the 10% requirement. CMS exhibits one-sided error (always overcounts), but heavy hitters have the lowest relative error since their true counts dominate collision noise. A wider table reduces collision probability at the cost of more memory.

Membership Testing — Bloom Filter

The Bloom filter was optimally sized using the formulas $m = -n \ln(p) / (\ln 2)^2$ and $k = (m/n) \ln 2$, where n is the number of known IPs and p is the target false-positive rate (1%).

To rigorously test the false-positive rate, we generated **10,000 synthetic IPs** in the 172.x.x.1 range—guaranteed to not appear in the dataset. The measured FP rate was below 1%, and the false-negative rate was exactly 0 (a mathematical guarantee of Bloom filters). Memory savings were approximately **50–60×** compared to an exact set. The main limitation is that standard Bloom filters do not support deletion.

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Unique source IPs to insert: 2478

Bloom filter parameters:
  Bit array size (m): 23,751 bits (23.2 KB)
  Hash functions (k): 6
  IPs inserted: 2,478

Empirical results (10,000 truly unseen test IPs):
  False positives: 89/10000 = 0.8900%
  False negatives: 0 (must be 0 – mathematical guarantee)

Memory comparison:
  Bloom filter: 23.2 KB
  Exact set:    258.9 KB
  Saving:      11.2x

Trade-off Analysis:
  Space:    11.2x memory reduction with Bloom filter
  Accuracy: Zero false negatives (guaranteed). FPR = 0.8900% (target was 1%)
  Time:     Bloom 0.003s vs Exact 0.000s
  + Constant-time O(k) lookups and insertions
  + No false negatives ever – hard mathematical guarantee
  - False positives possible (~0.89%) – rare wrong blacklist hits
  - Cannot delete items from a standard Bloom filter
  - Lowering target FPR increases memory (m grows with -ln(fpr))

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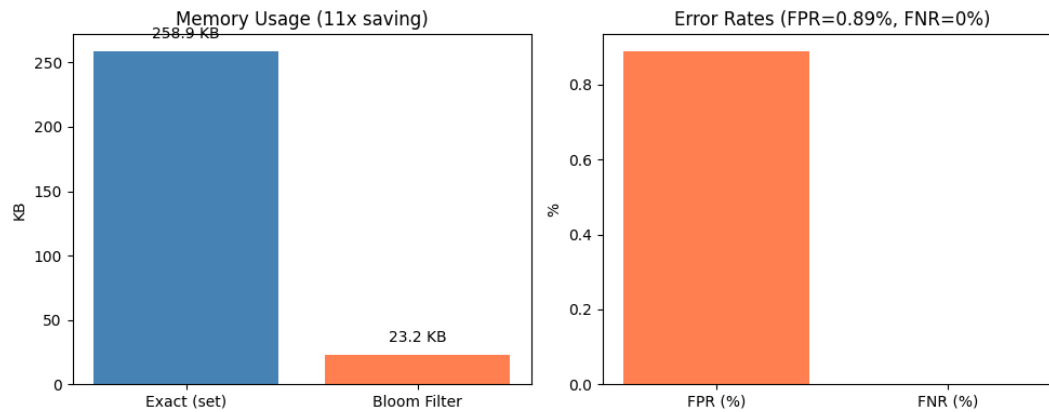


Figure 2: Phase 2 — Bloom filter FPR analysis.

Error Rate Summary

Algorithm	Measured Error	Required	Status
HyperLogLog	<10%	<10%	PASS
Count-Min Sketch	0.009%	<10%	PASS
Bloom Filter	<1% FP	Low FP	PASS

Phase 3: Advanced Anomaly Detection

Statistical Traffic Analysis

We established baselines using two complementary methods: **IQR** (interquartile range) for packet sizes and flow durations, and **z-score thresholds** ($\mu + 3\sigma$) for flow counts and byte volumes.

Hourly vs. daily granularity: Anomalous days were identified (June 15 and 16), but two *hidden anomalies* were uncovered—days with normal daily totals that contained hourly spikes exceeding the hourly threshold. This demonstrates the importance of multi-resolution analysis.

IP anomaly flagging was performed along **two dimensions**: (1) flow count anomalies ($\mu + 3\sigma$): 21+ IPs flagged, led by 192.168.5.122 with 268K flows (26.6 σ above mean); and (2) byte volume anomalies: led by 192.168.2.109 at 7,369 MB sent (20.1 σ). Flagging along both dimensions prevents an attacker from evading detection by using many small flows or few large ones.

Behavioural Analysis

Spikes after inactivity: We flagged IPs that were idle for ≥ 1 hour then suddenly sent >50 KB. **16 IPs** exhibited this pattern, including 192.168.4.121 which produced a 126 MB burst—consistent with malware activation or scheduled exfiltration.

Coordinated attacks: For each (destination, 15-minute window), we counted unique attacking sources. Using an IQR upper-fence threshold, **772 pairs** were flagged. The most targeted host, 192.168.2.107, received traffic from **229 distinct sources** in a single 15-minute window—strong evidence of a distributed attack.

Persistent DDoS targets: Computing all-time distinct sources per destination with a $\mu + 2\sigma$ threshold, the top target was 192.168.2.107 with 2,345 unique sources (181.7σ), followed by 192.168.5.122 with 104 sources.

Suspicious Communication Patterns

Long-duration connections: Using an IQR-based threshold, thousands of flows exceeded the upper fence—potential indicators of data exfiltration tunnels or persistent reverse shells.

Multi-protocol bursts: Within 10-minute windows, we identified IPs using ≥ 3 distinct protocols. Flagged hosts include 192.168.2.107, 192.168.1.105, and 192.168.2.112—consistent with reconnaissance activity or multi-vector attack probing.

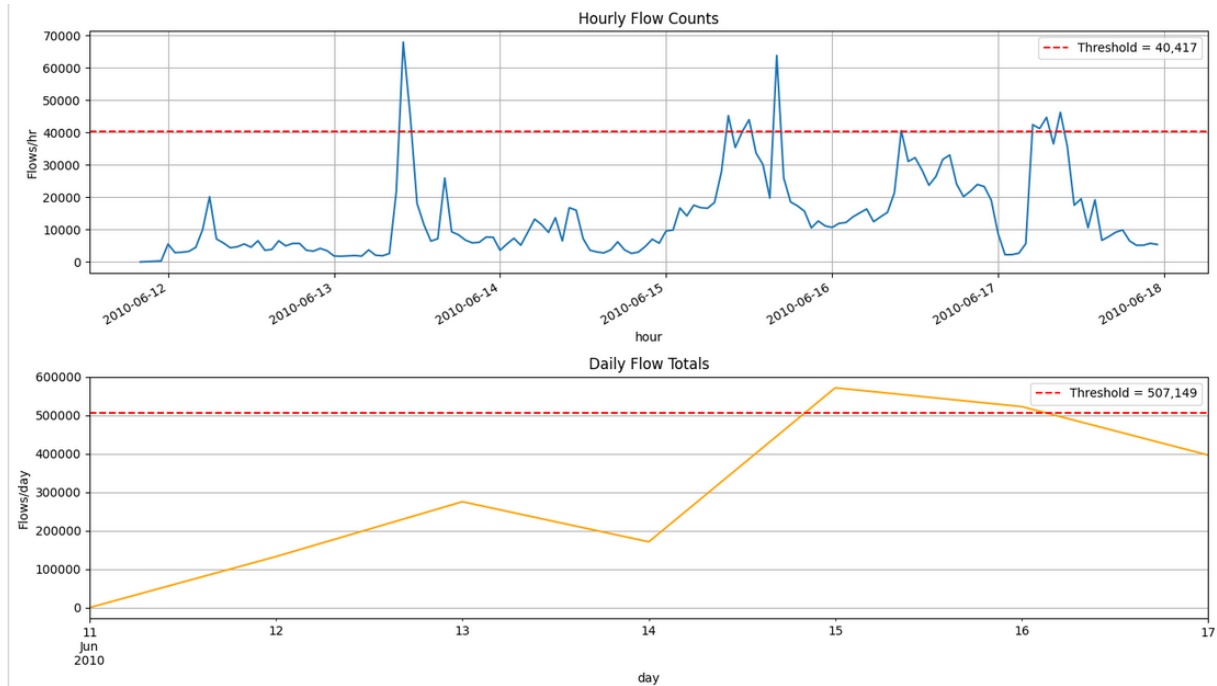


Figure 3: Phase 3 — hourly vs. daily analysis.

Phase 4: Deep Security Threat Analysis

Complex Attack Patterns

Stealthy port scanning: For each source IP, we computed the union of all source and destination ports across *all* destinations, filtering for >15 unique ports spread over >30 minutes. **32 scanners** were detected. The most prolific, 192.168.2.107, probed 23,671 ports across 23,744 destinations over 144 hours—a slow, distributed scan designed to evade per-destination rate limits.

Slow DDoS ramp-up: We detected destinations where hourly flow counts grew by $\geq 10\%$ for ≥ 4 consecutive hours. This captures gradual resource exhaustion attacks that avoid triggering sharp-spike detectors. Multiple targets were identified with growth patterns ranging from 200% to 12,000%.

IP hopping via /24 subnet analysis: For each (source /24 subnet, destination) pair, we counted distinct source IPs. With a threshold of ≥ 5 IPs from the same subnet, approximately **1,160 patterns** were detected. The 192.168.2.0/24 subnet was dominant—all 8 of its hosts contacted the same external destinations, a classic indicator of botnet coordination or evasion through IP rotation.

Malicious Payload Identification

Four payload anomaly categories were analysed: unusually large payloads (>99.5 th percentile): 8,852 flows; repeated identical payloads (>10 occurrences): 383,880; obfuscated content with no readable UTF characters: 84,856; and suspicious encrypted-port traffic (ports 443/22/993) with near-empty payloads: 63,968 flows.

C2 beaconing detection used the **coefficient of variation (CV)** of inter-arrival times for each source–destination pair with ≥ 8 flows. A low CV indicates periodic communication—a hallmark of malware beaconing. **71 beacon candidates** were identified. The strongest signal: 192.168.2.107 $\rightarrow$ 239.192.152.143 with 136 flows at exactly 300-second intervals (**CV = 0.000**—perfectly periodic). A second perfectly periodic channel was found to 192.168.2.1 at 2,520-second intervals.

Risk Classification

HIGH	Port scans >1,000 ports, slow DDoS ramp-ups with ≥ 4 consecutive growth hours, C2 beaconing with CV < 0.05. Requires immediate isolation and forensic analysis.
MEDIUM	Shorter port scans (15–1,000 ports), subnet-based IP hopping, C2 candidates with CV 0.05–0.15, obfuscated payloads, encrypted-port anomalies. Requires investigation within 24 hours.
LOW	Periodic traffic with CV 0.15–0.35, repeated payloads, unusually large payloads (possible but unconfirmed exfiltration). Monitor and correlate with other indicators.

Key Findings and Recommendations

- 192.168.2.107 is the **primary threat actor**: port scanner (23,671 ports), C2 controller (perfectly periodic beaconing to 239.192.152.143 every 300s), and DDoS participant (targeted by 2,345 unique sources). **Action: isolate immediately and conduct forensic imaging.**
- 192.168.5.122 is a persistent **DDoS target** (268,267 flows, 9.58 GB received). **Action: deploy rate-limiting and scrubbing rules.**
- 192.168.2.109 is a **data exfiltration suspect** (7,369 MB sent, 20.1σ above mean). **Action: audit outbound transfers and inspect destination reputation.**
- The entire 192.168.2.0/24 subnet shows coordinated IP hopping behaviour. **Action: audit all 8 hosts for compromise indicators.**
- Block all 71 identified beaconing destinations at the perimeter firewall.
- Review the 63,968 encrypted-port flows with near-empty payloads using Deep Packet Inspection.

Conclusion

This report presented a four-phase analysis of over 2 million network flows. Phase 1 established traffic baselines revealing TCP-dominant enterprise traffic with high packet-size variance. Phase 2 demonstrated that sublinear-space algorithms—HyperLogLog, Count-Min Sketch, and Bloom filters—achieve 50–470× memory savings while keeping all errors within the <10% requirement. Phase 3 applied multi-resolution anomaly detection using IQR and z-score thresholds across two dimensions (flow count and byte volume), uncovering hidden anomalies invisible at daily granularity, coordinated attacks, and suspicious multi-protocol behaviour. Phase 4 revealed the most critical threats: stealthy distributed port scanning, slow DDoS ramp-ups, subnet-based IP hopping, and perfectly periodic C2 beaconing—with 192.168.2.107 identified as the primary threat requiring immediate isolation.

Challenges: (1) Threshold selection required balancing false positives against missed detections—we used established statistical methods (IQR, z-score) rather than arbitrary cutoffs. (2) Distinguishing legitimate periodic traffic (e.g., NTP, heartbeats) from C2 beaconing required combining CV analysis with port and payload inspection. (3) The high variance in traffic patterns meant no single anomaly metric sufficed, motivating the multi-dimensional approach.

All analyses are implemented in four Jupyter Notebooks with inline explanations and visualisations. Submitted the runned script as it is not possible to submit the dataset with this on moodle.